Introduzione al corso

Corso di "Trasmissione dei segnali digitali"
Laurea Magistrale in Ingegneria Informatica

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Introduzione al corso

- Riferimenti docente
  - Donatella Darsena, DIETI
  - Palazzina 2 (Via Claudio)
  - Tel: 081-7683156, E-mail: darsena@unina.it
  - Orario di ricevimento: giovedì 14:30-16:30 (su piattaforma Teams)
  - Eventuali appuntamenti fuori orario: da concordare via e-mail
- Libri di testo
  - Upamanyu Madhow, "Fundamentals of Digital Communication", Cambridge, 2008
  - Per alcuni argomenti saranno indicati altri testi di riferimento
  - Appunti delle lezioni
- Modalità di svolgimento del corso ed esame
  - Le lezioni saranno svolte impiegando slide e lavagna
  - La prova finale prevede un colloquio orale
  - Le date di esame saranno disponibili sul sito docenti di UNINA

Principali argomenti del corso

- Rappresentazione complessa in banda base dei segnali
- Processi aleatori
- Modulazione digitale dei segnali
- Stima dei parametri e sincronizzazione
- Comunicazione digitale su canali selettivi in frequenza
- Elementi fondamentali di teoria dell'informazione e codifica di canale

The Communication Age

- We define *communication* as information (i.e., voice, audio, video, data files, web pages, etc.) transfer between different points in space or time (e.g., storage).
- Today, communication enters our daily lives in different ways.
- The telephones as well as mobile smart phones and devices at our hands, the radios and televisions in our living rooms, or the computer terminals with access to the Internet in our offices and homes are all capable of providing rapid communications from every corner of the globe.
- Communication provides the senses for ships on the high seas, aircraft in flight, and rockets and satellites in space.
- Communication through a wireless telephone keeps a car driver in touch with the office or home miles away, no matter where.
- Communication provides the means for social networks to engage in different ways (texting, speaking, visualizing), whereby people are brought together around the world.
- Communication keeps a weather forecaster informed of conditions measured by a multitude of sensors and satellites.

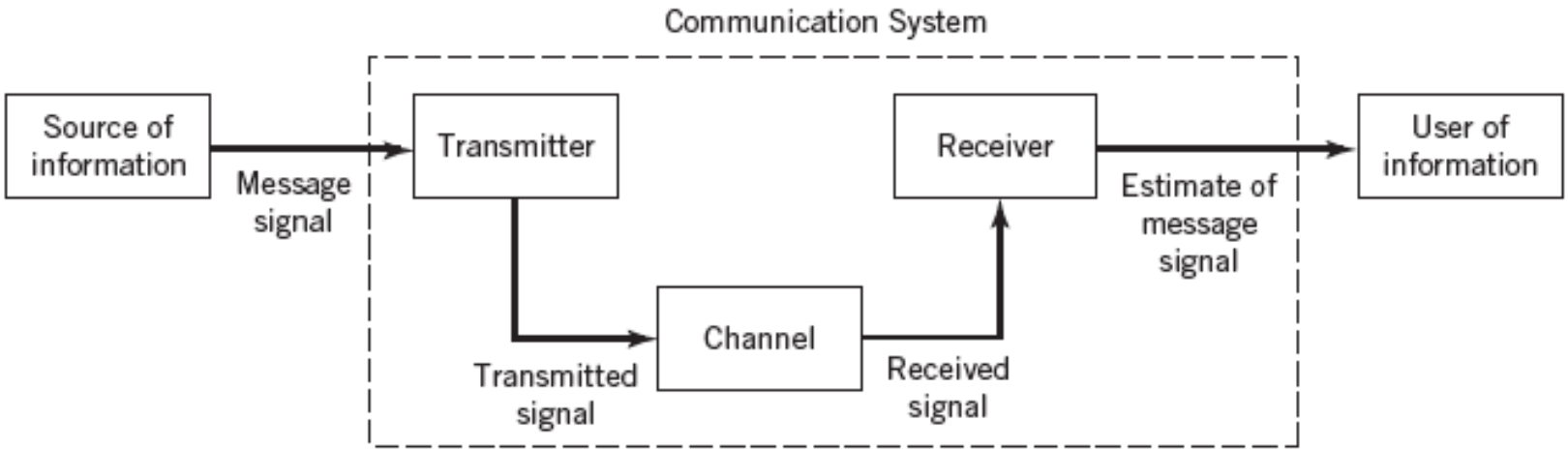
The Communication Process (1/2)

- Communication involves implicitly the transmission of information from one point to another through:
  - generation of a message signal - voice, music, image, or data;
  - description of that message signal with a certain measure of precision, using a set of symbols - electrical, aural, or visual;
  - encoding of those symbols in a suitable form for transmission over a physical medium;
  - transmission of the encoded symbols to the desired destination;
  - decoding and reproduction of the original symbols;
  - re-creation of the original message signal with some definable degradation in quality, the degradation being caused by unavoidable imperfections in the system.

The Communication Process (2/2)

- There are, of course, many other forms of communication that do not directly involve the human mind in real time.
  - For example, in computer communications involving communication between two or more computers, human decisions may enter only in setting up the programs or commands for the computer, or in monitoring the results.
- Irrespective of the form of communication process being considered, there are three basic elements to every communication system, namely,
  - transmitter;
  - channel;
  - receiver.
- The cascade of the transmitter and receiver is also known as “transceiver”.

Elements of a communication system



Transmission of information

- The transmitter is located at one point in space, the receiver is located at some other point separate from the transmitter, and the channel is the physical medium that connects them together as an integrated communication system.
- The purpose of the transmitter is to convert the message signal produced by the source of information into a form suitable for transmission over the channel.

Reception of information

- As the transmitted signal propagates along the channel, it is distorted due to channel imperfections. Moreover, noise and interfering signals (originating from other sources) are added to the channel output: the received signal is a corrupted version of the transmitted signal.
- The receiver has the task of operating on the received signal so as to reconstruct a recognizable form of the original message signal for an end user or information sink.
- The communication process in every communication system, irrespective of its kind, is statistical in nature.

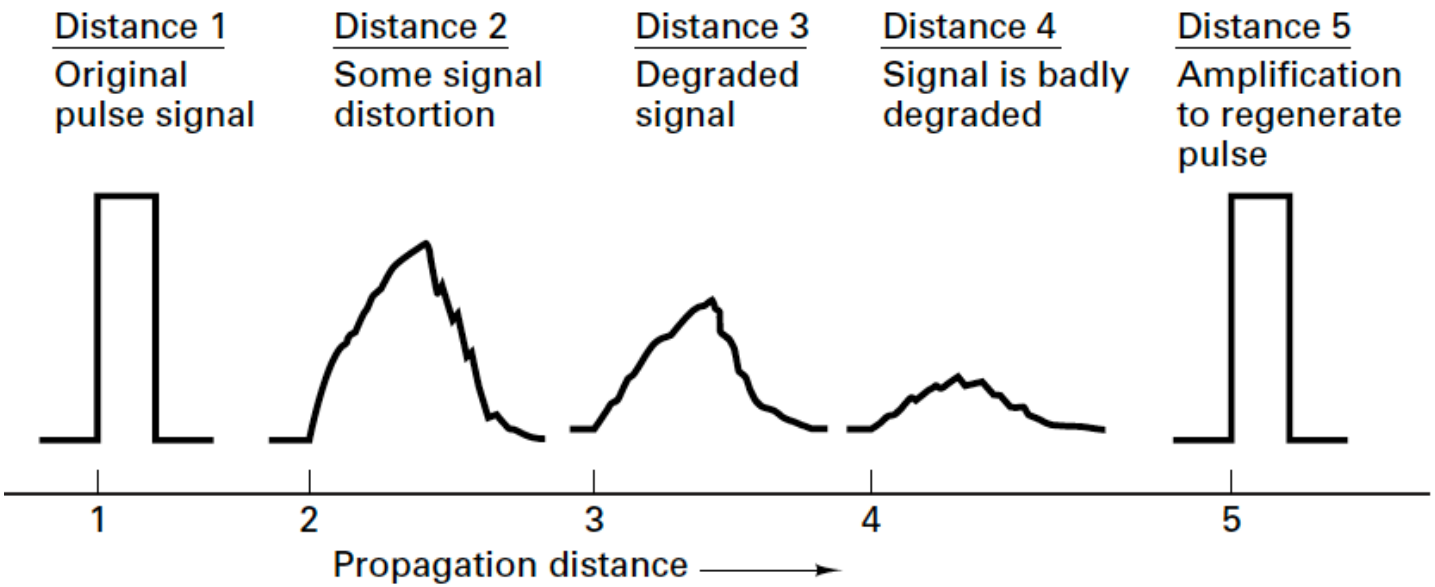
Digital communication

- In digital communication, the information being transferred is represented in digital form (most commonly, binary digits or *bits*).
- In analog communication, information takes on a continuum of values.
- Why are communication systems "going digital"?
- The primary advantage is the ease with which digital signals, compared with analog signals, are regenerated.
- With digital techniques, extremely low error rates producing high signal fidelity are possible through error detection and correction but similar procedures are not available with analog.

Pulse degradation and regeneration (1/2)

Remark

An ideal binary digital pulse propagating along a transmission line.



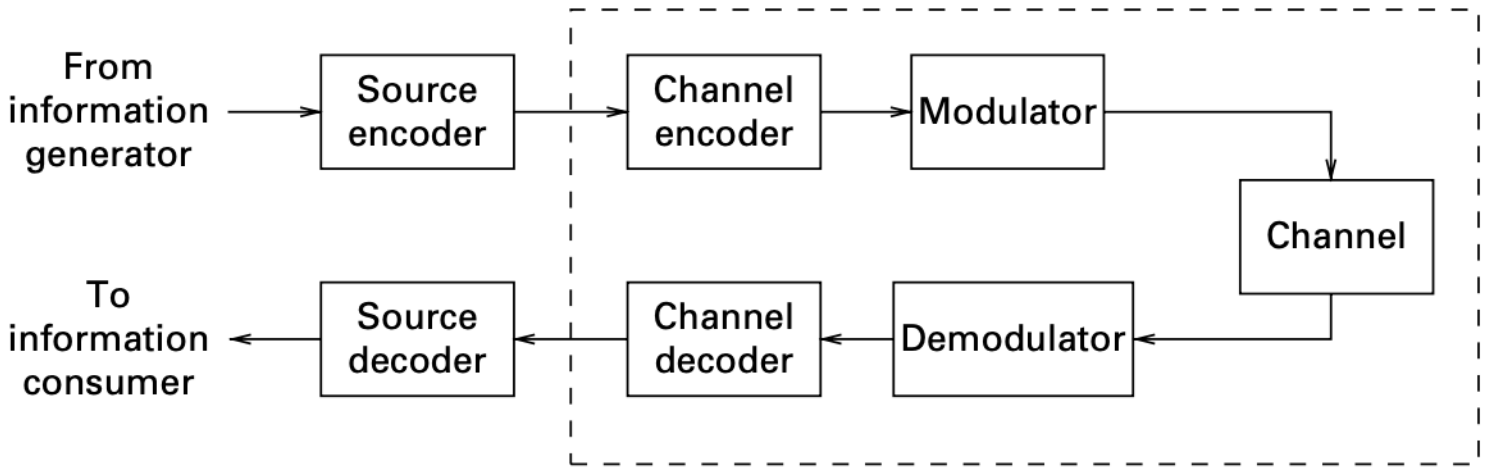
Pulse degradation and regeneration (2/2)

- The shape of the waveform is affected by two mechanisms:
  - As all transmission lines and circuits have some non-ideal frequency transfer function, there is a distorting effect.
  - Unwanted electrical noise or other interference further distorts the pulse waveform.
- Both of these mechanisms cause the pulse shape to degrade as a function of line length.
- During the time that the transmitted pulse can still be reliably identified (before it is degraded to an ambiguous state), the pulse is amplified by a digital amplifier that recovers its original ideal shape $\Rightarrow$ the pulse is thus "reborn" or regenerated.
- Circuits that perform this function at regular intervals along a transmission system are called regenerative repeaters.

Other advantages of digital circuits

- Digital circuits are more reliable and can be produced at a lower cost than analog circuits.
- Digital hardware lends itself to more flexible implementation than analog hardware [e.g., microprocessors, digital switching, and large-scale integrated (LSI) circuits].
- Different types of digital signals (data, telegraph, telephone, television) can be treated as identical signals in transmission and switching - a bit is a bit.
- Digital messages can be handled in autonomous groups called packets. Digital techniques lend themselves naturally to signal processing functions that protect against interference and jamming, or that provide encryption and privacy.
- Much data communication is from computer to computer, or from digital instruments or terminal to computer. Such digital terminations are naturally best served by digital techniques.

Diagram of Digital Communication System



Source encoder

- According to information theory, any information can be efficiently represented in digital form up to arbitrary precision, with the number of bits required for the representation depending on the required fidelity.
- Source coding produces analog-to-digital (A/D) conversion (for analog sources) and reduces the redundancy in the original information in a manner that takes into account the end user's requirements.
- For example, voice can be sent at 64 kbit/s for conventional wireline telephony, MP3 players for consumer applications may employ typical bit rates of 128 kbit/s, while high-end digital audio studio equipment may require around ten times higher data rates.

Channel encoder and modulator

- Channel encoder introduces redundancy in a controlled fashion in order to combat errors that may arise from channel imperfections and noise.
- The output of the channel encoder is a codeword, from a channel code, which is properly designed for the channel characteristics and the requirements dictated by higher network layers.
- For example, error detection plus a request for retransmission for delay-insensitive applications or error correction for real-time applications.
- The modulator translates message symbols or channel symbols (when channel coding is used) into analog waveforms that can be transmitted over the physical channel.

Channel

- The channel distorts the transmitted waveform in a manner that may significantly differ in different settings.
- For wireline communication, for example, the channel is well modeled as a linear time-invariant (LTI) system and thus described in terms of its impulse response.
- For wireless mobile communication, the channel may vary because of the relative mobility between the transmitter and receiver.
- Since wireless is a broadcast medium, multiple-access interference due to simultaneous transmissions can arise.

Thermal noise

- At the receiving side, the received signal is corrupted by thermal noise generated by electronic components.
- A conductive element with two terminals may be characterized by its resistance R (ohm).
- This resistive or lossy element contains free electrons that have some random motion if the resistor has a temperature above absolute zero.
- This *random* motion causes a noise voltage to be generated at the terminals of the resistor.
- Although the noise is small, when the noise is amplified by a high-gain receiver it can become a problem.
- Thermal noise can be characterized resorting to quantum mechanics.

Demodulator and channel decoder

- The demodulator processes the analog received waveform which is a distorted and noisy version of the transmitted waveform.
- One of its key tasks is *synchronization*: the demodulator must account for the fact that the channel can produce phase, frequency, and time shifts, and that the clocks and oscillators at the transmitter and receiver are not synchronized a priori.
- Another task of the demodulator may be *channel equalization*.
- The ultimate goal of the demodulator is to produce decisions on the transmitted symbols to be fed to the channel decoder (*detection*).
- The channel decoder exploits the redundancy inserted at the transmitting side to produce an estimate of the information symbols that were the input to the channel encoder.

Source decoder

- The source decoder converts the estimated information bits produced by the channel decoder into a format that can be used by the end user.
- The final format may or may not be the same as the original format.
- For example, the original source encoder could have translated speech into text and then encoded it into bits and the source decoder may then display the text to the end user, rather then trying to reproduce the original speech.

Key advantages of the digital communication approach

- Source-independent design
  - Once the information is transformed into the bits by the source encoder, it can be stored or transmitted *without interpretation*.
  - The storage or communication medium can be independent of the source characteristics: a variety of information sources can share the same communication medium.
- Channel-optimized design
  - For each communication link, the channel encoder/decoder and modulator/demodulator can be optimized for the specific channel characteristics.
  - Since the bits are regenerated at each link, there is no "noise accumulation".
- Two remarks:
  - This framework is based on separation of source and channel coding. The *separation principle* yields practical advantages and is optimal for point-to-point links (from information theory). In certain cases, joint source and channel coding can yield superior performance.
  - *Semantic communications* is an emerging topic that is receiving a wide range of attention. In a conventional communication system, the source is converted into bit sequences to process, which is accurately recovered at the receiver. Semantic communication, instead, transmit semantic meaning of the source: semantic coding captures the semantic features, depending on tasks or actions to be executed at the receiver.